Name – Shruti Hanmant Nangare

Batch – 1

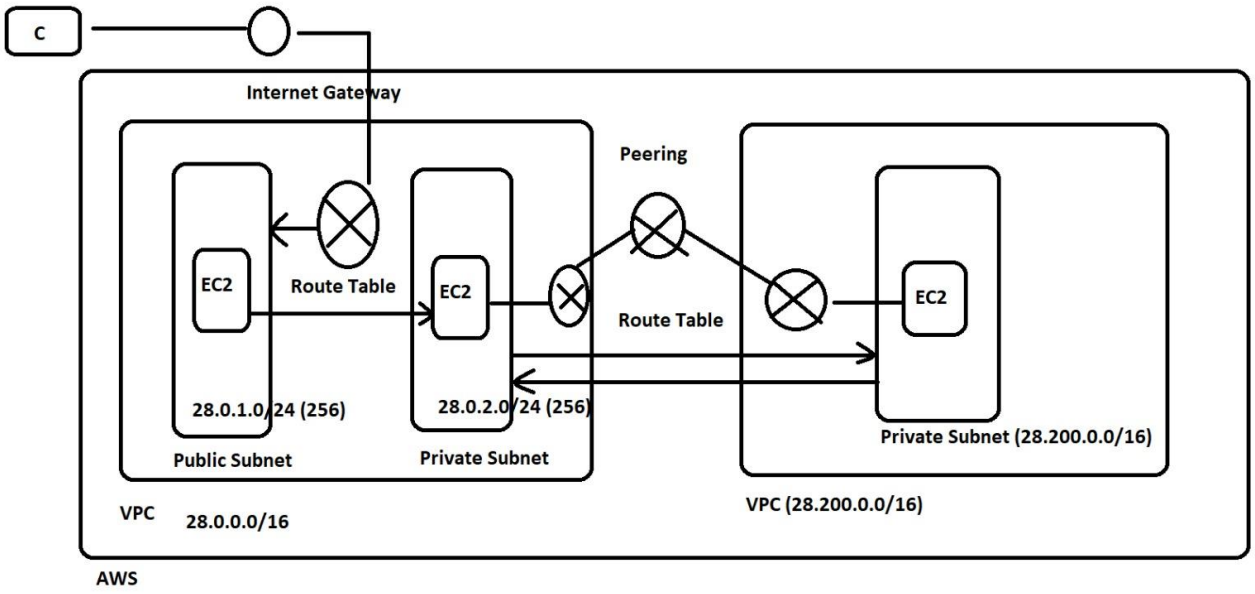
Project Title – AWS VPC Peering

Region - Mumbai

Completion Date – 29/10/2026

AWS VPC PEERING CONNECTION PROJECT

❖ ARCHITECTURE DIAGRAM

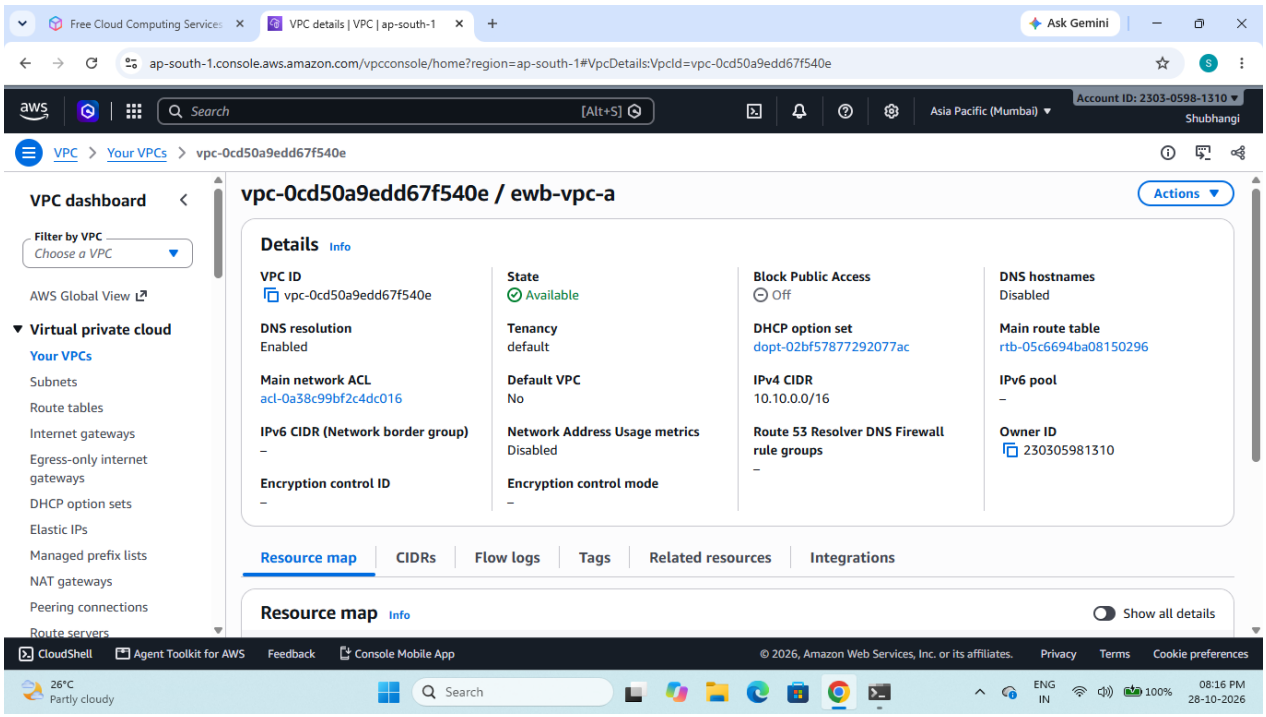


❖ RESOURCES INVENTORY

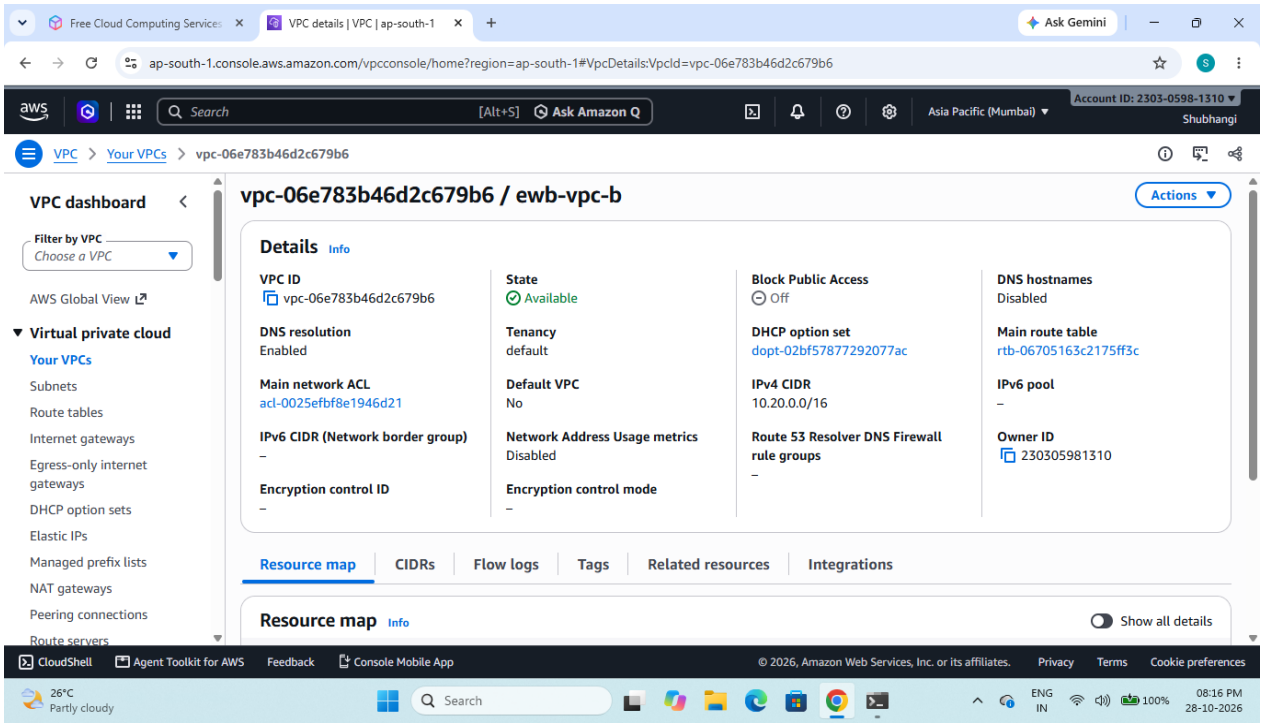
	Resources	Name
1.	VPC A	ewb-vpc-a
2.	VPC B	ewb-vpc-b
3.	Public subnet	ewb-a-public
4.	Private subnet A	ewb-a-private
5.	Private subnet B	ewb-b-private
6.	Internet Gateway	ewb-igw-a
7.	Public Route Table	ewb-rt-a-public
8.	Private Route Table A	ewb-rt-a-private
9.	Private Route Table B	ewb-rt-b-private
10.	Bastion SG	ewb-sg-bastion
11.	App-A SG	ewb-sg-app-a
12.	App-B SG	ewb-sg-app-b
13.	Peering	ewb-a-b-peer
14.	EC2 Instance	ewb-bastion, ewb-app-a, ewb-app-b

❖ VPC

- VPC A

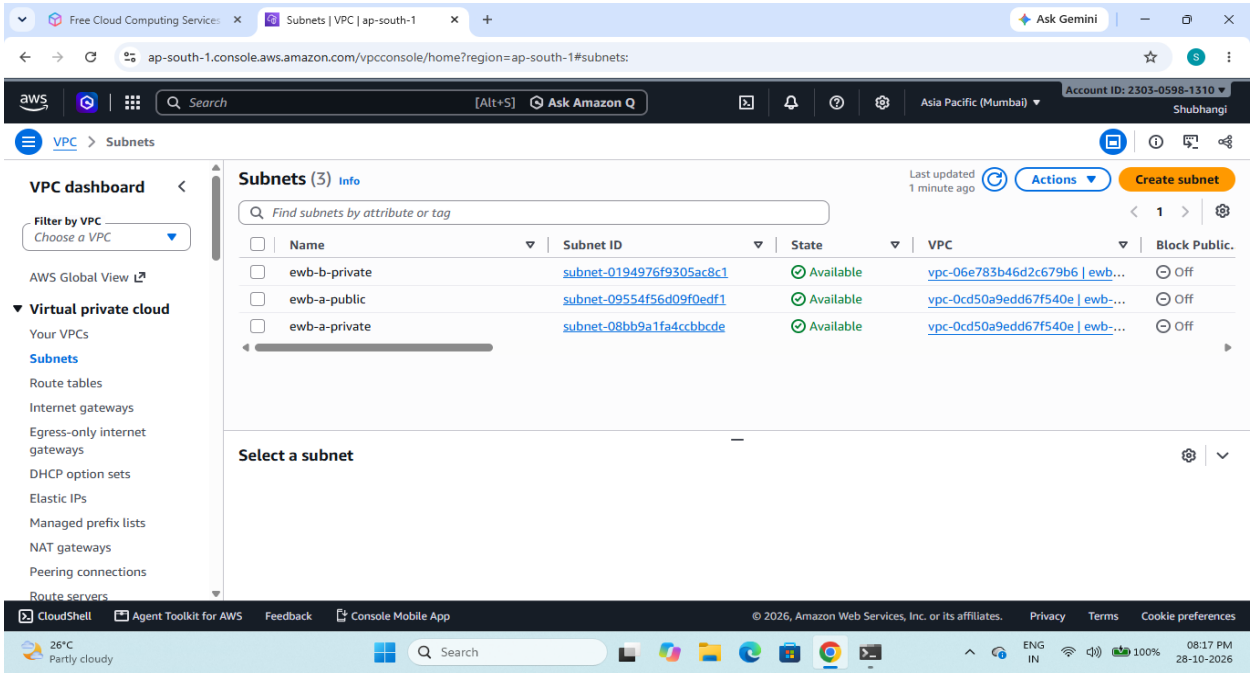


- VPC B

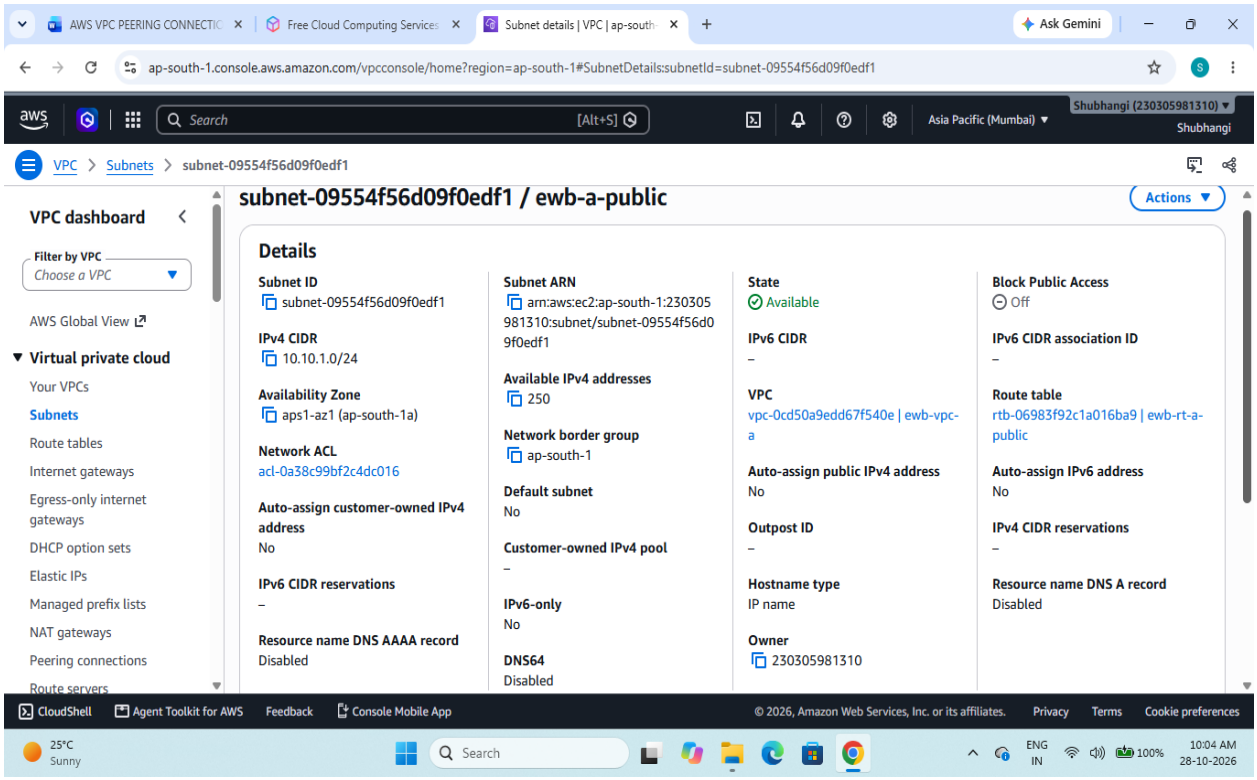


❖ SUBNETS

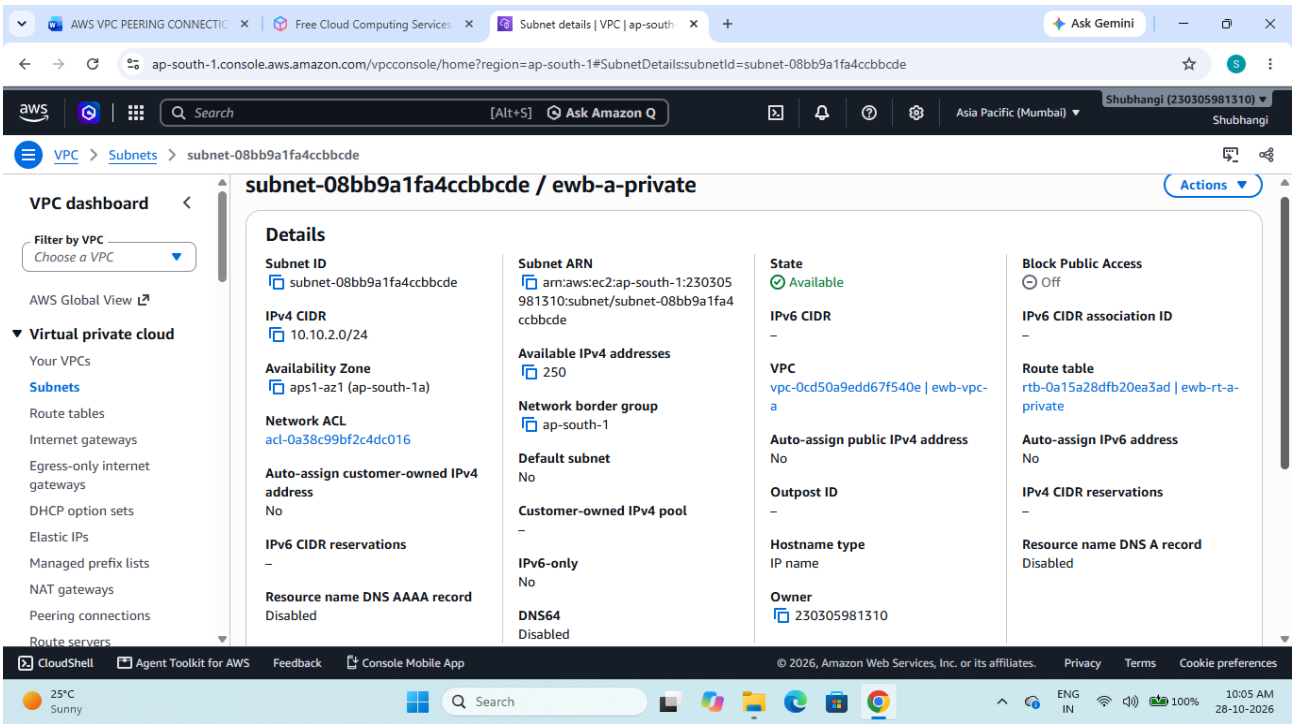
- ALL SUBNETS



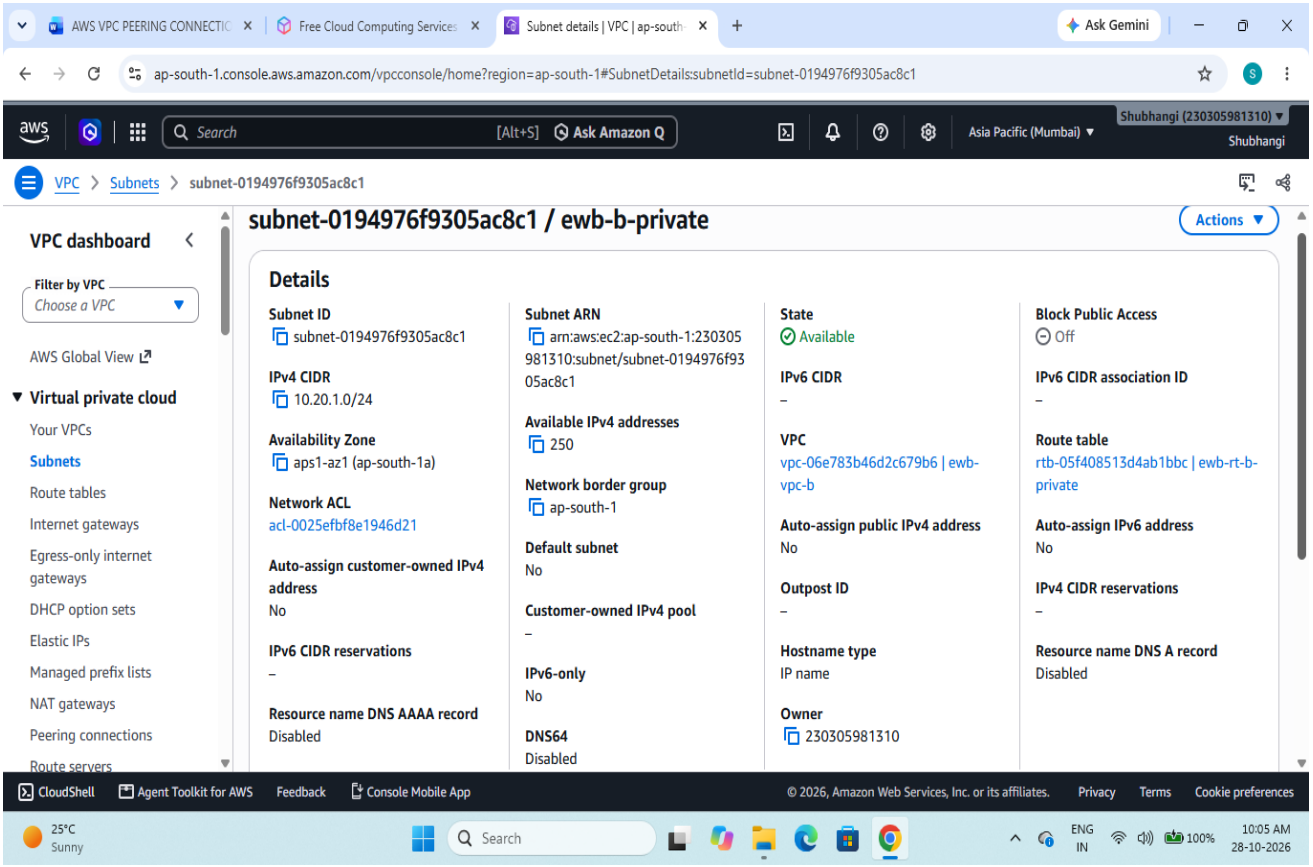
- SUBNET – ewb-a-public



- SUBNET – ewb-a-private

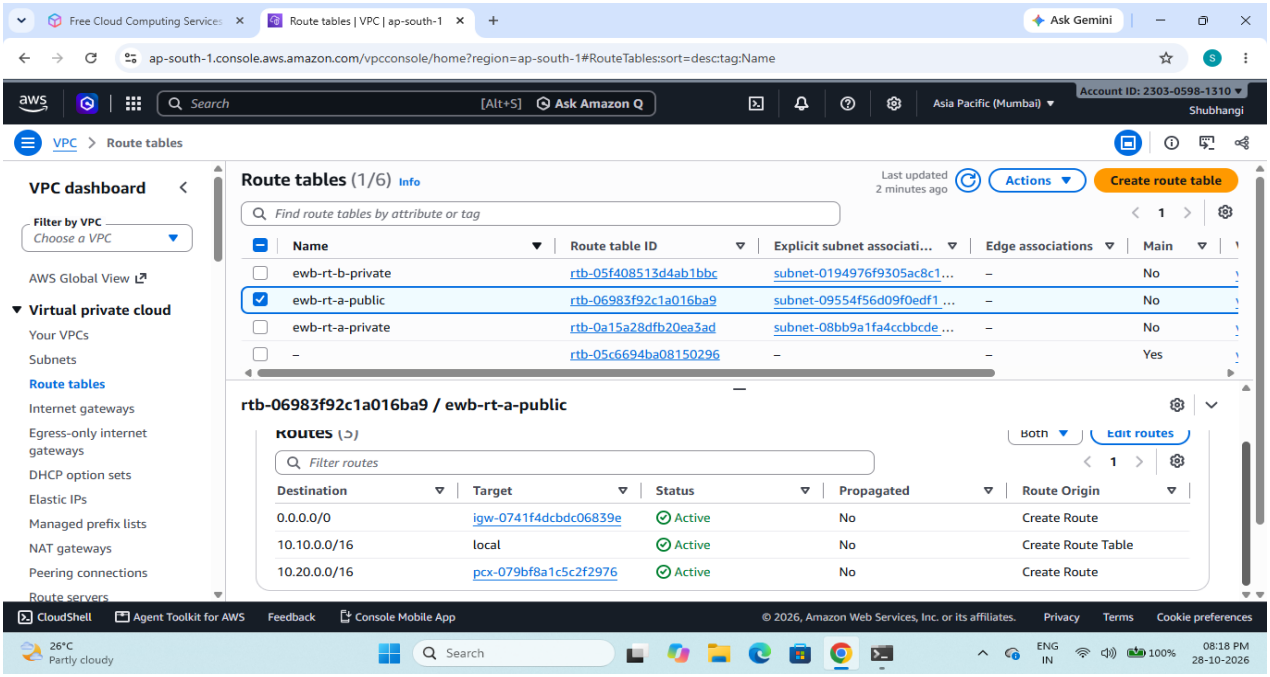


- SUBNET – ewb-b-private

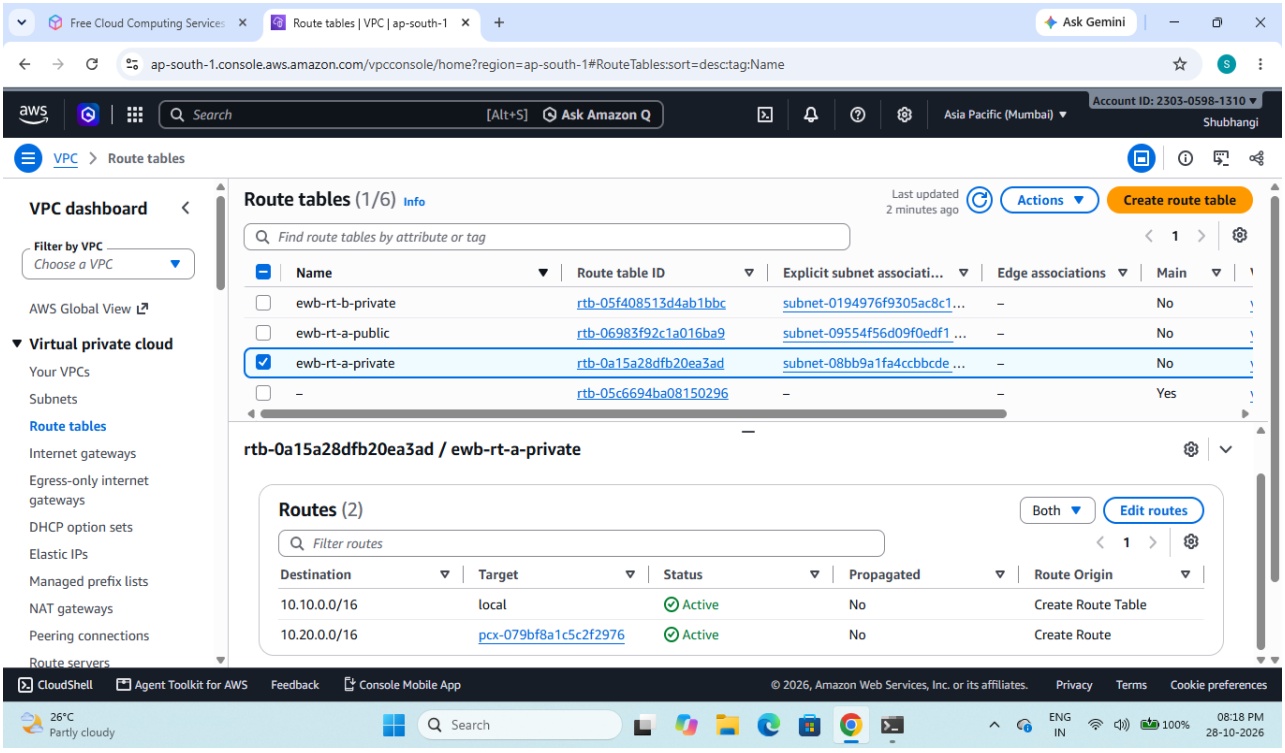


❖ ROUTE TABLES

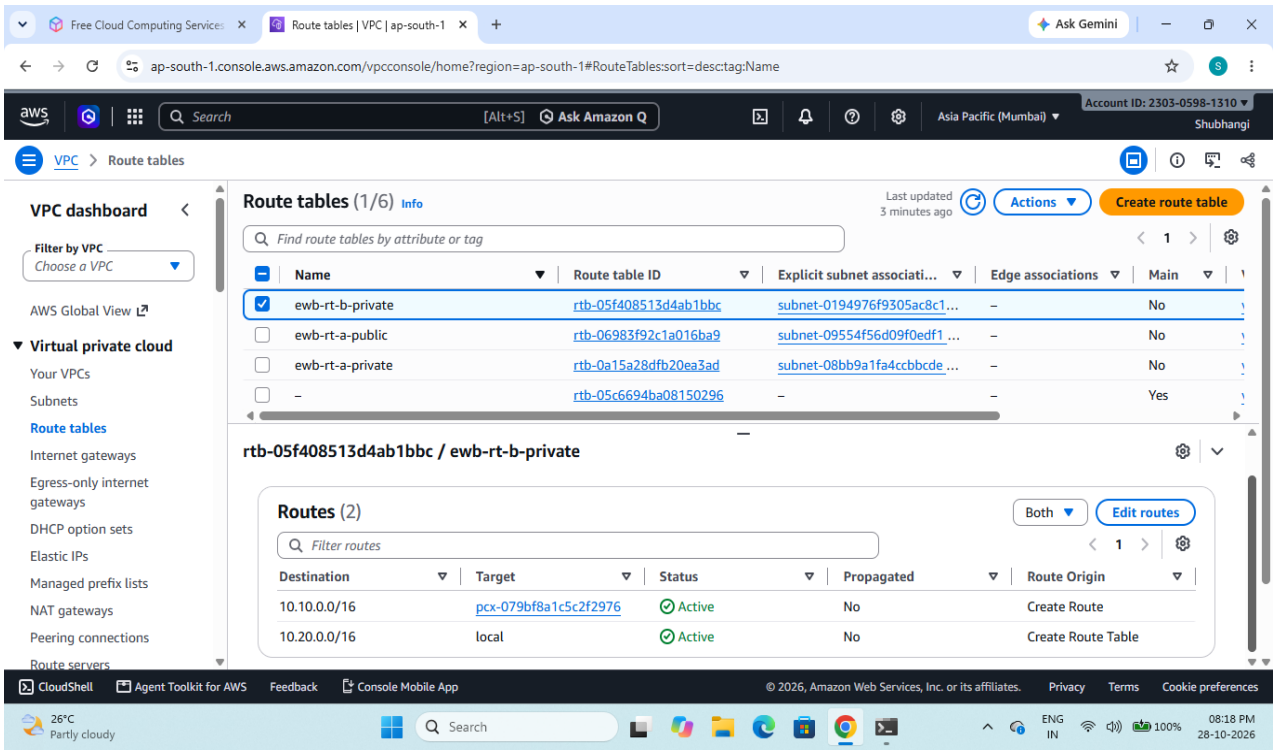
- Route table – ewb-rt-a-public



- Route table – ewb-rt-a-private

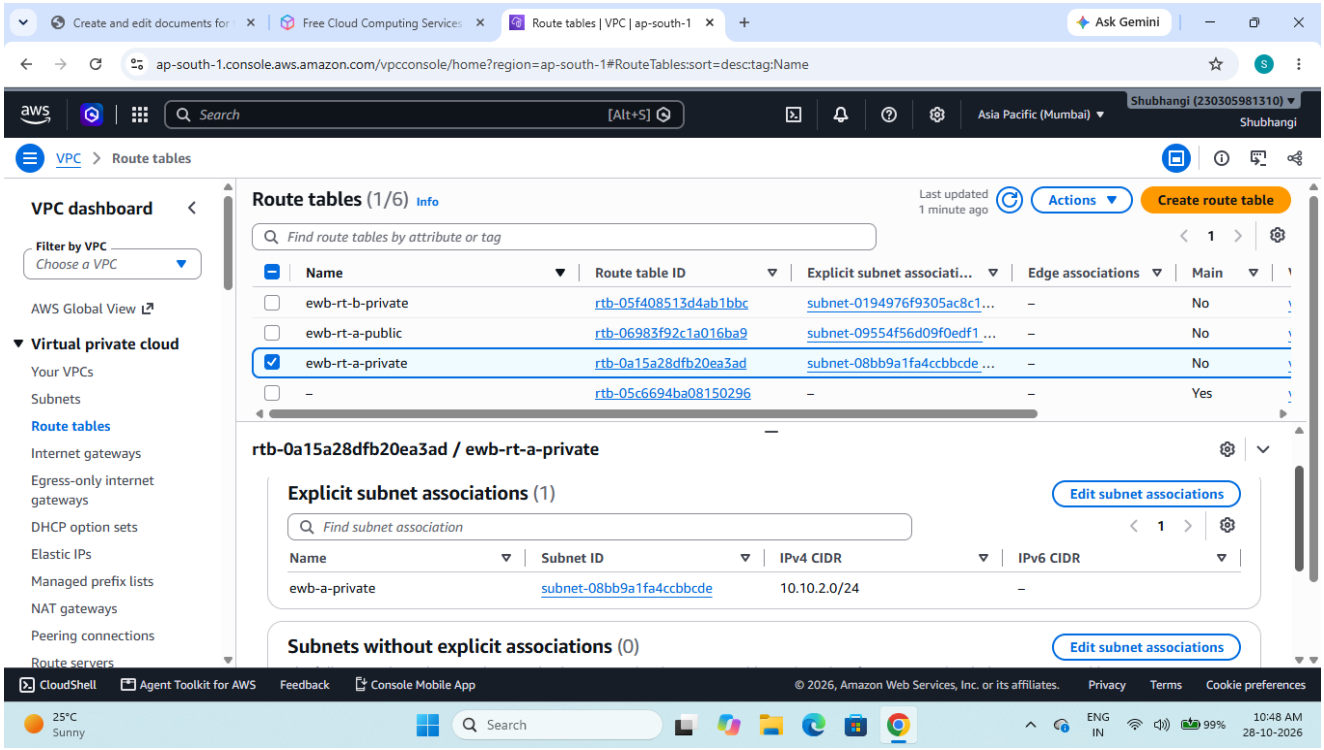


- Route table – ewb-rt-b-private

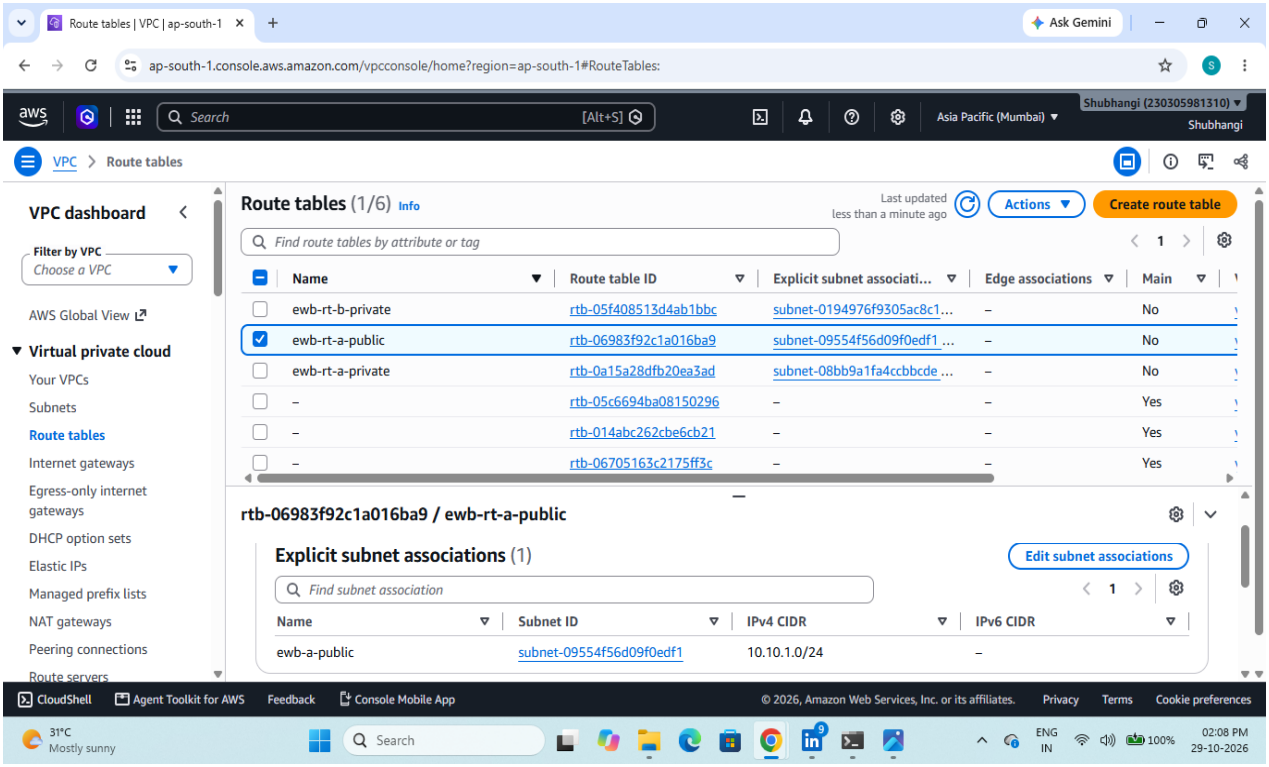


❖ SUBNET ASSOCIATION

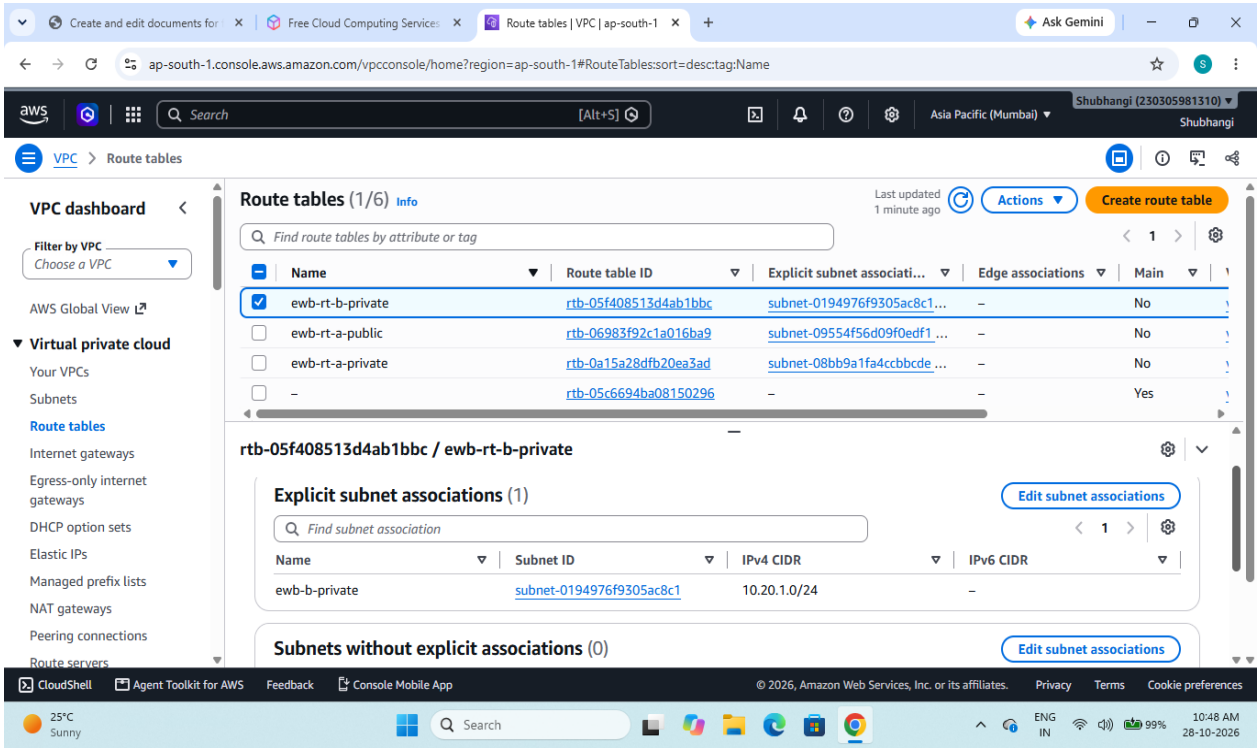
- Subnet Association – ewb-rt-a-private



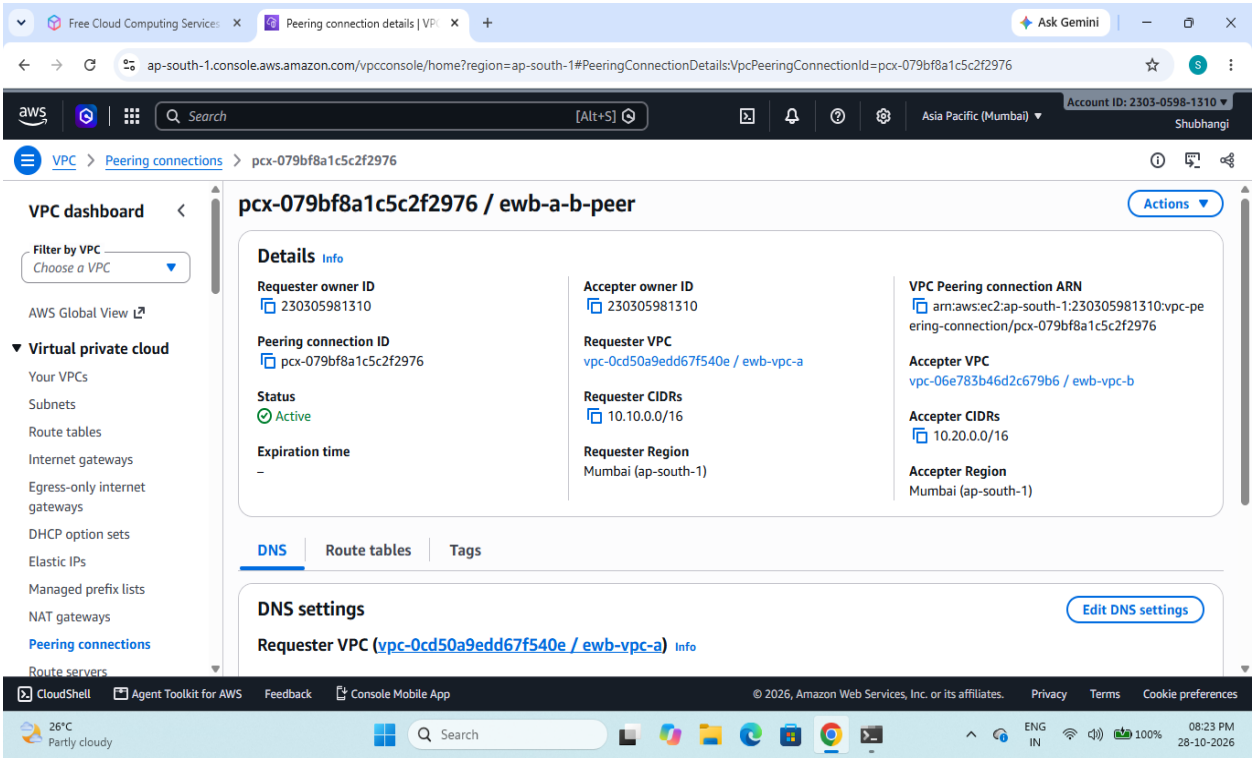
- Subnet Association – ewb-rt-a-public



- Subnet Association – ewb-rt-b-private

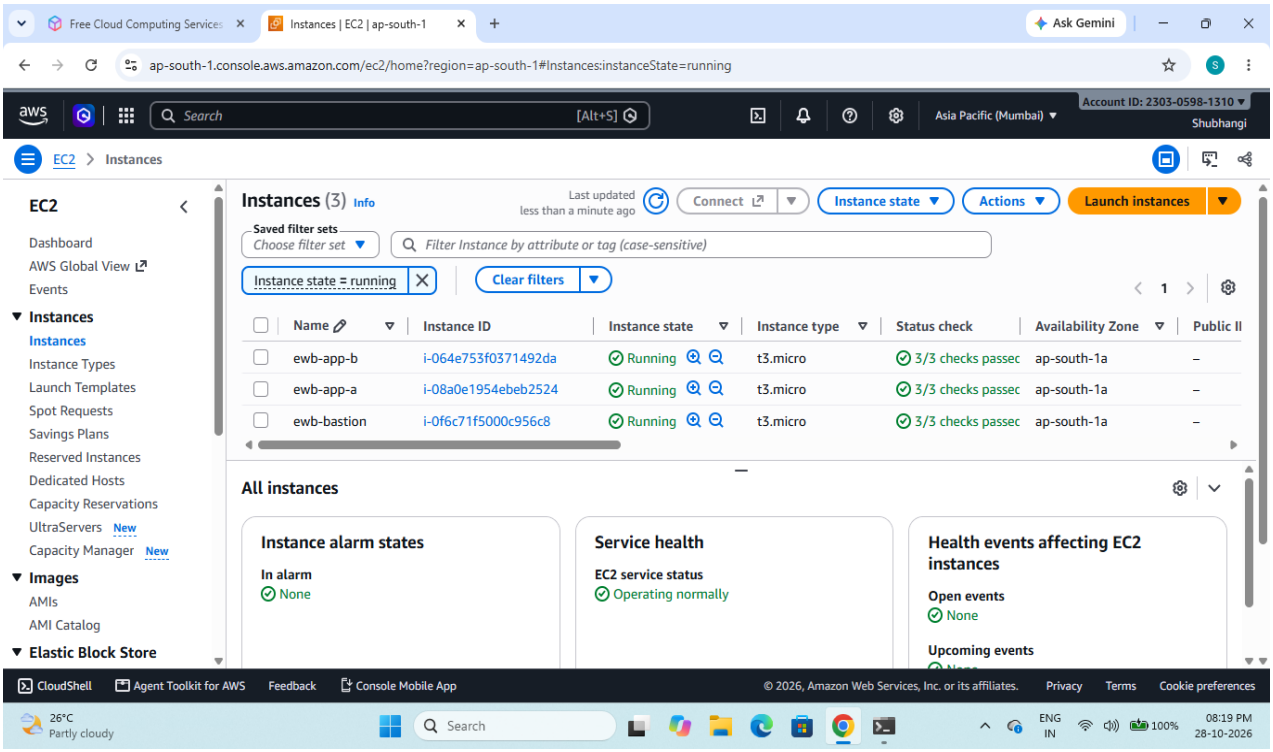


❖ VPC PEERING

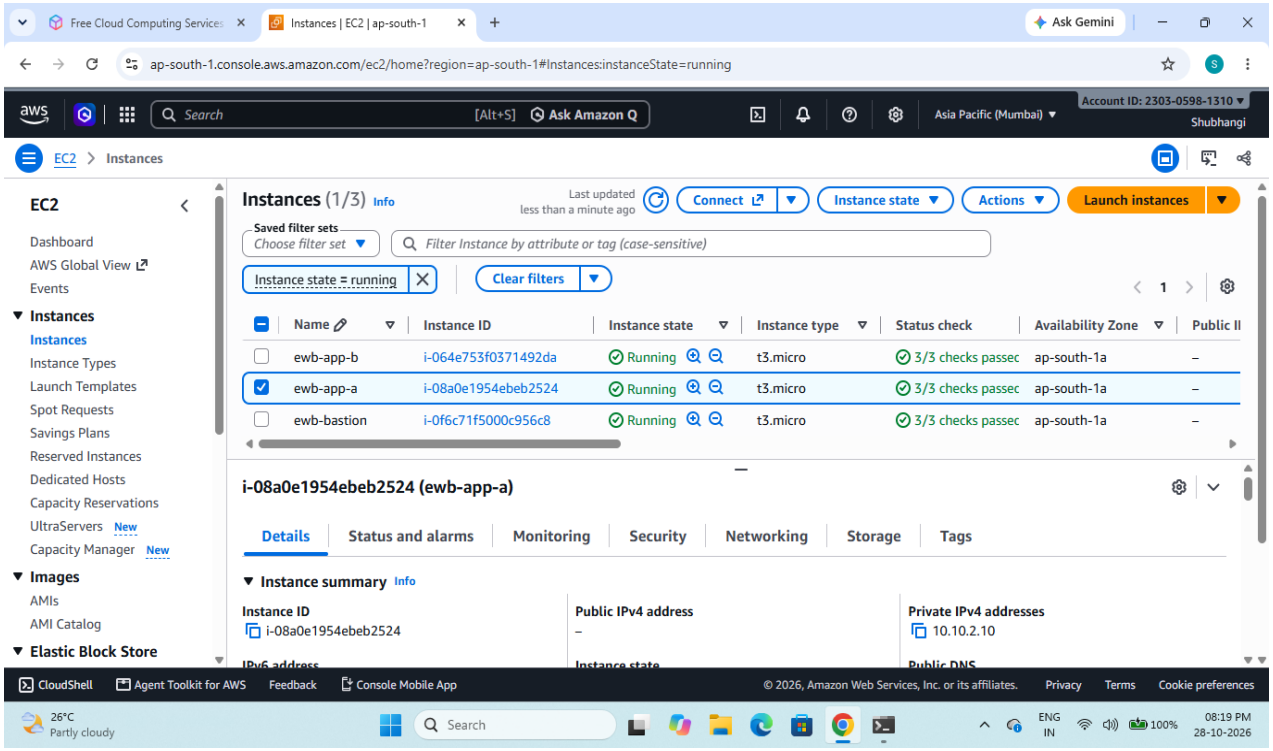


❖ EC2 INSTANCES

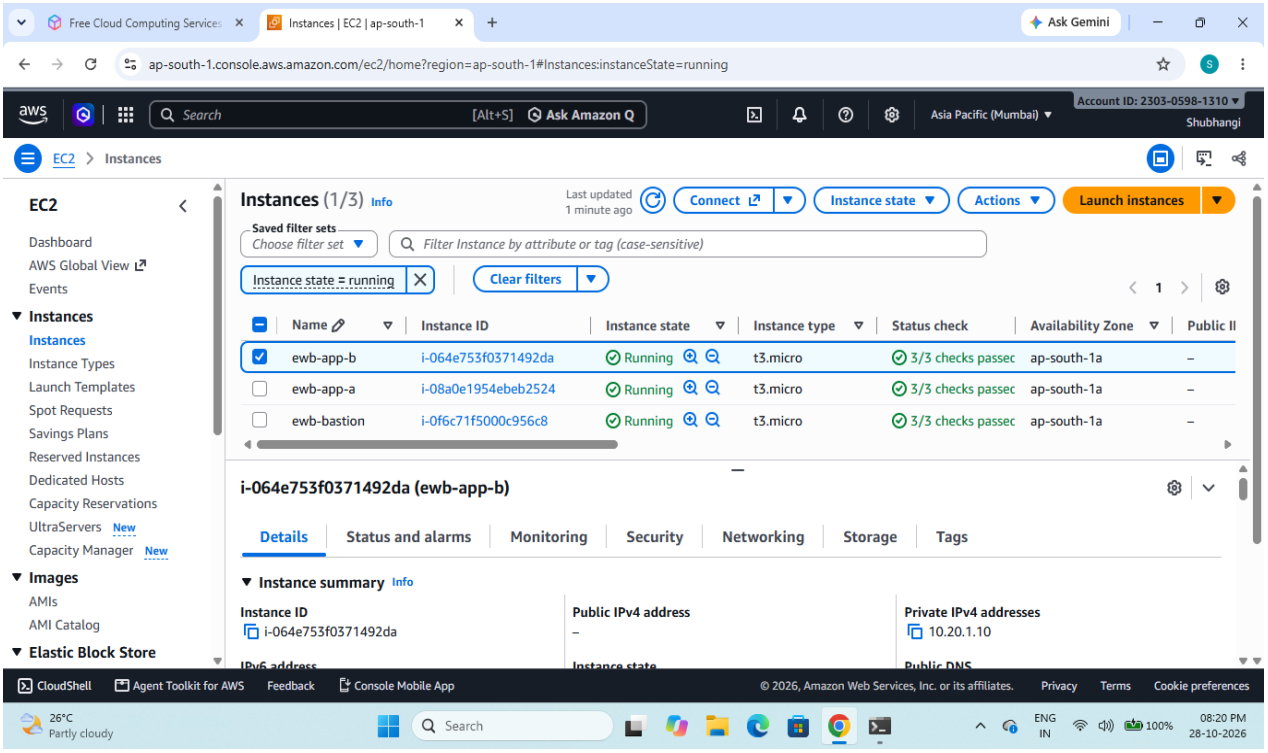
- EC2 ALL INSTANCES



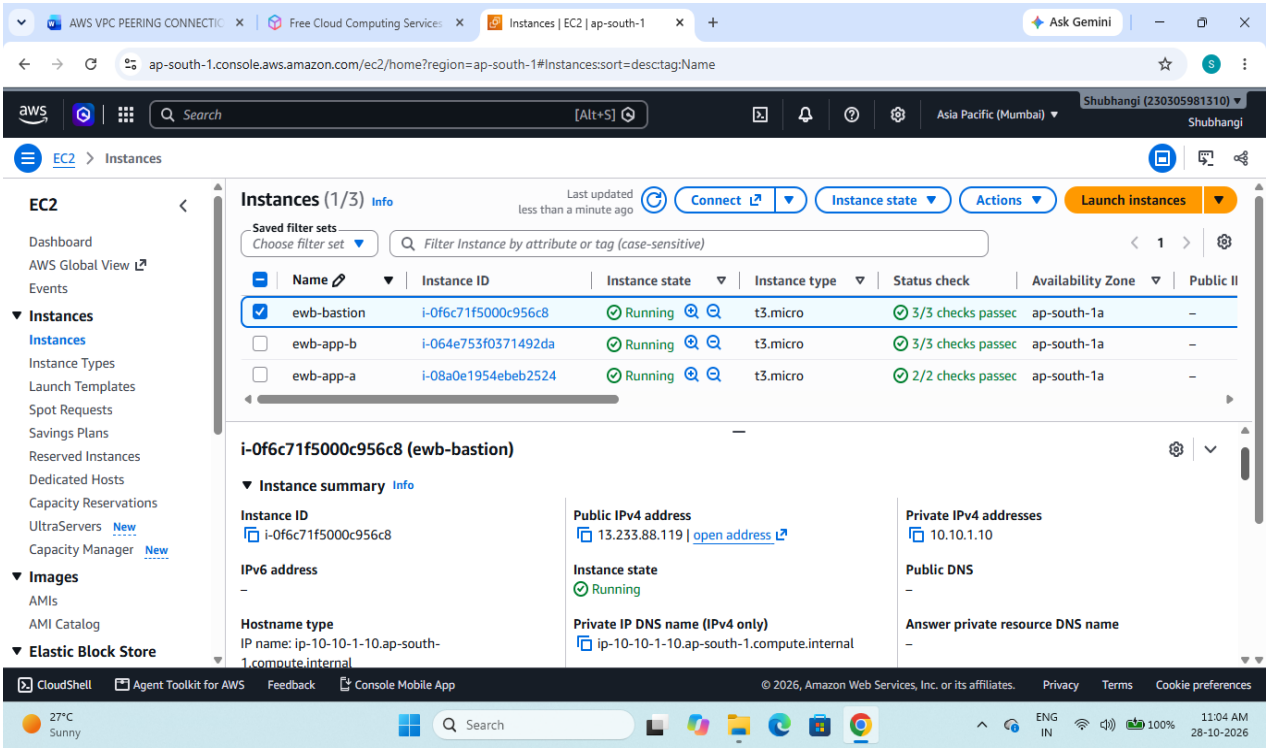
- EC2 INSTANCE – ewb-app-a



- EC2 INSTANCE – ewb-app-b

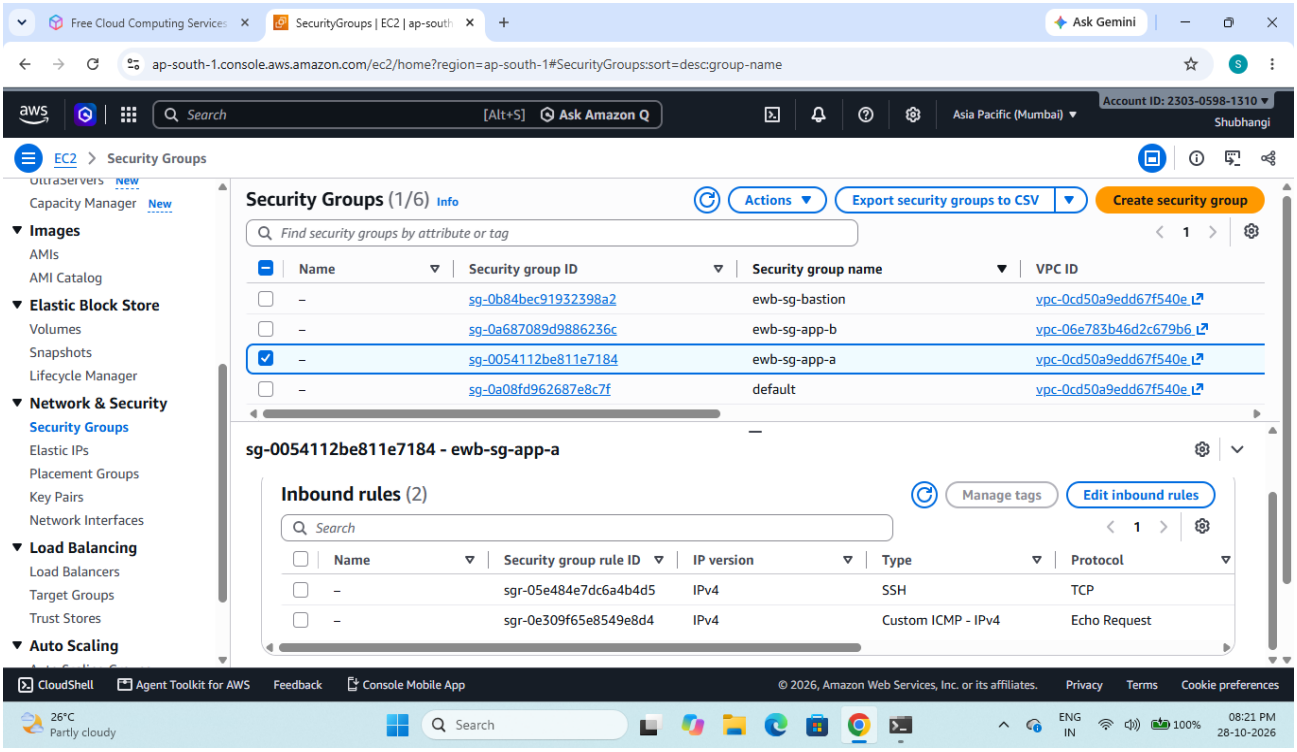


- EC2 INSTANCE – ewb-bastion

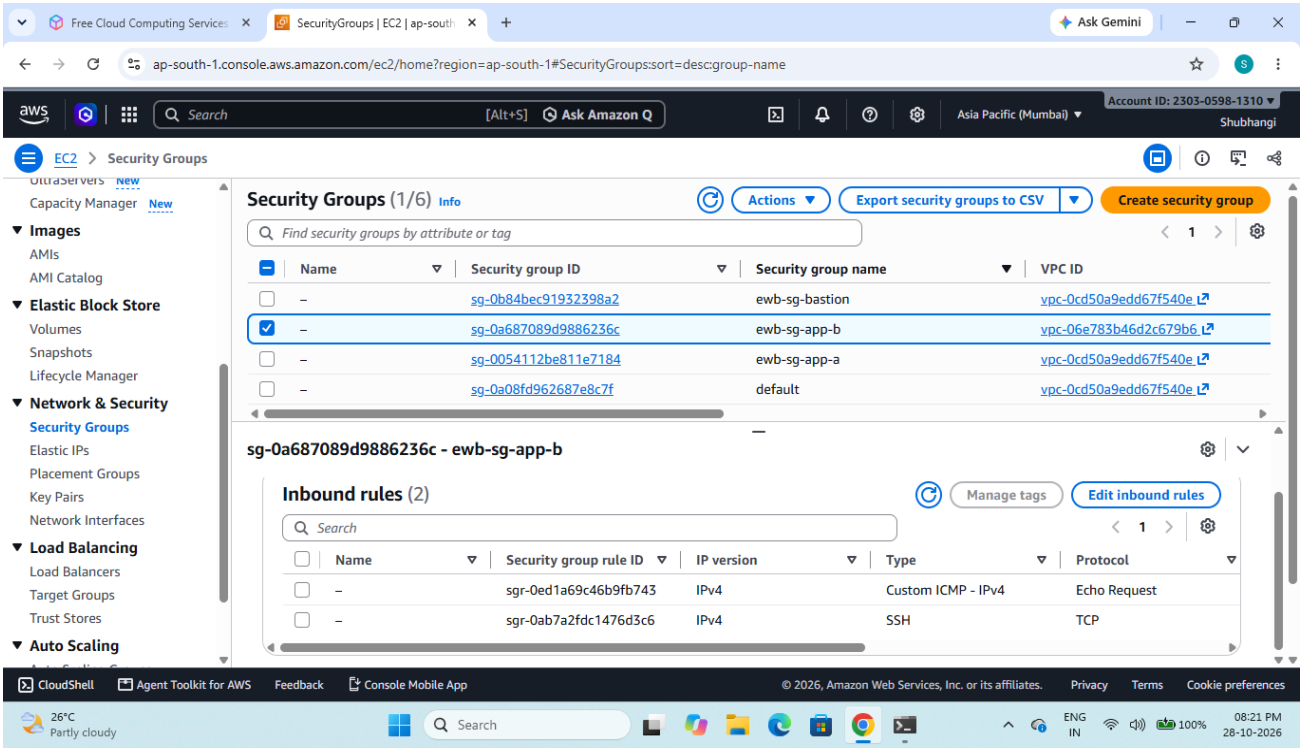


❖ SECURITY GROUPS

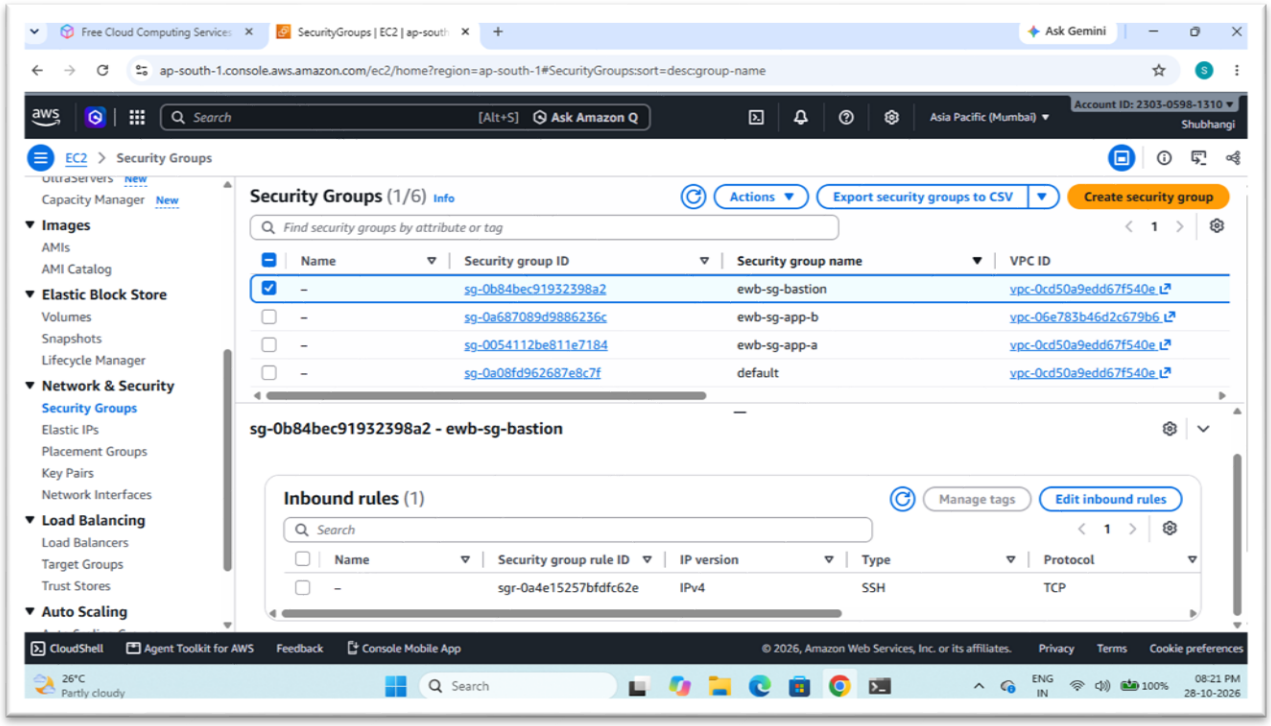
- SECURITY GROUP – ewb-sg-app-a



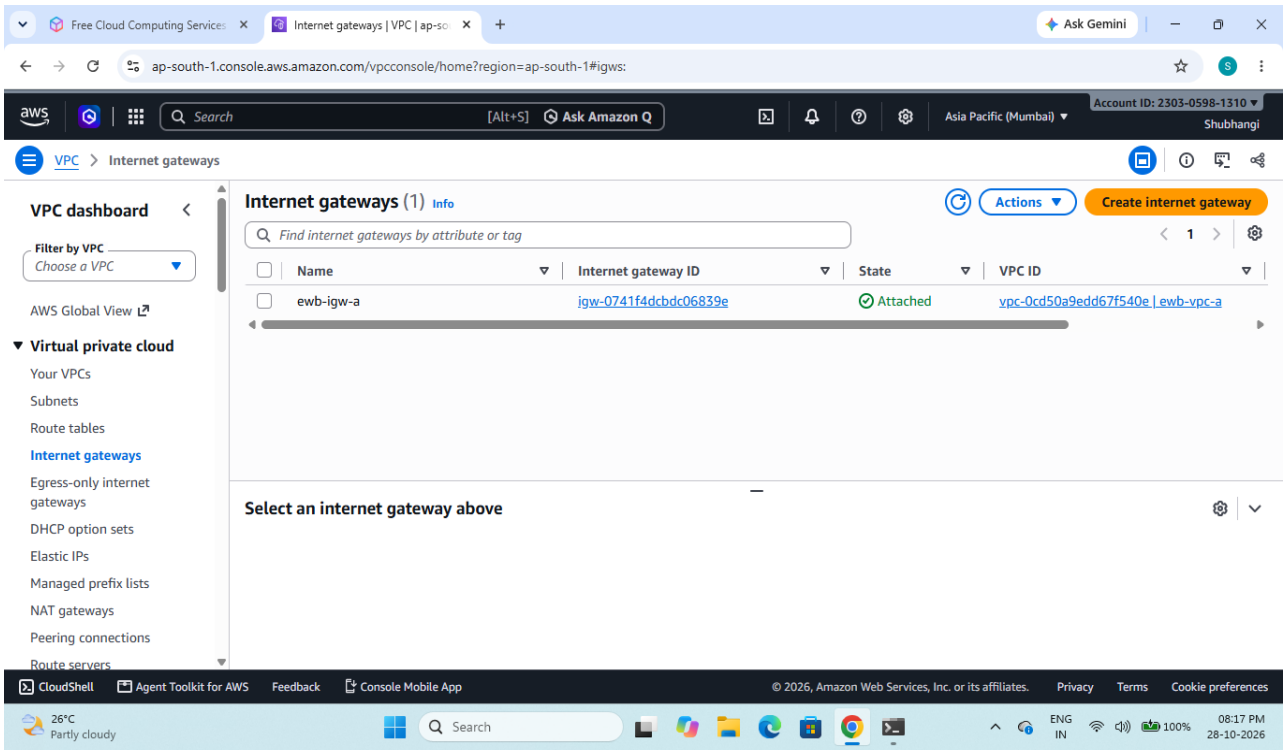
- SECURITY GROUP – ewb-sg-app-b



- SECURITY GROUP – ewb-sg-bastion

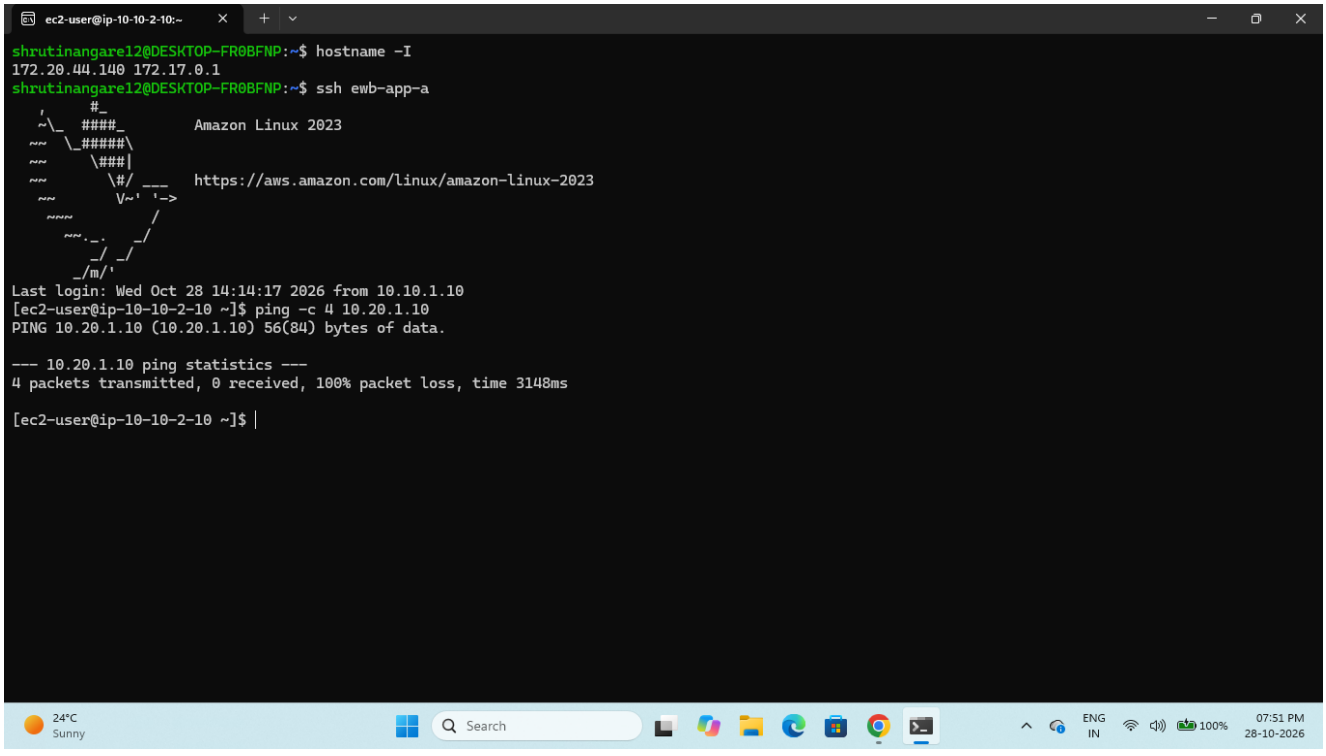


❖ INTERNET GATEWAY

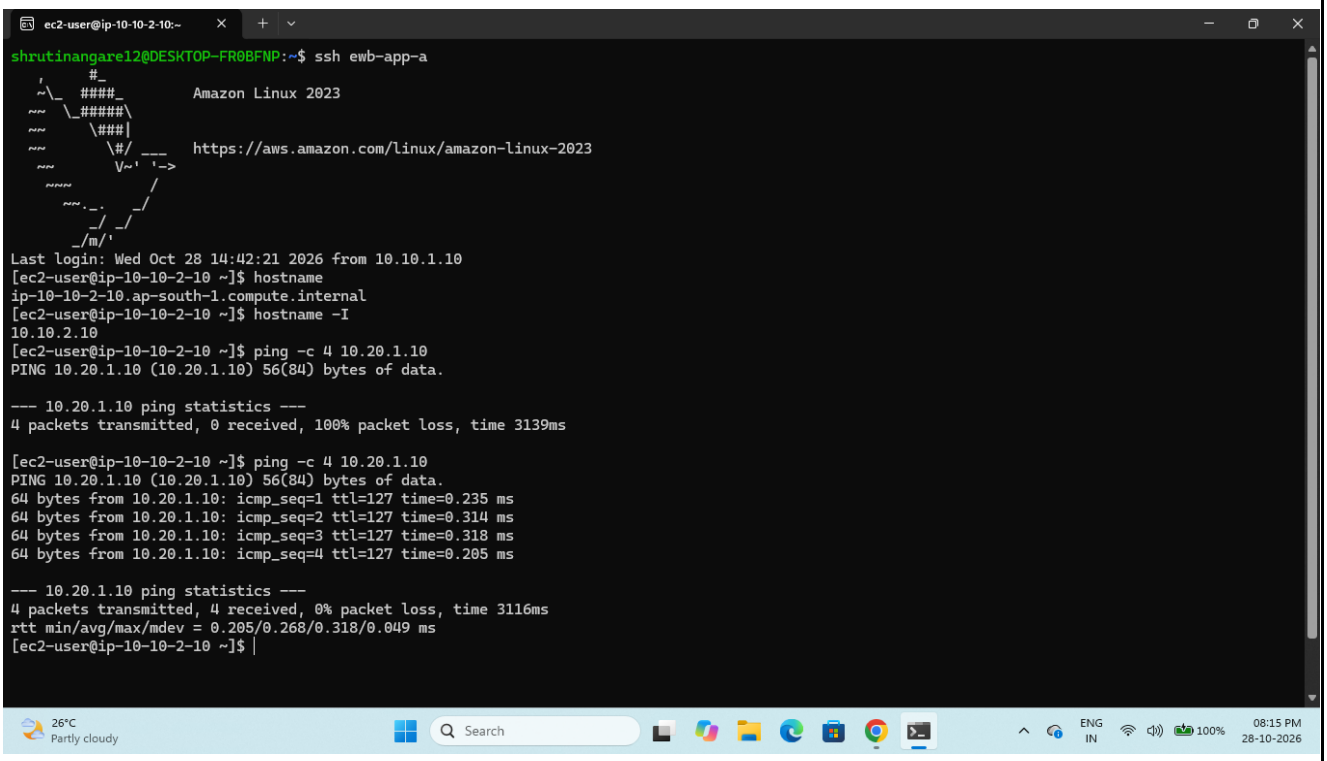


❖ BASELINE FAILURE

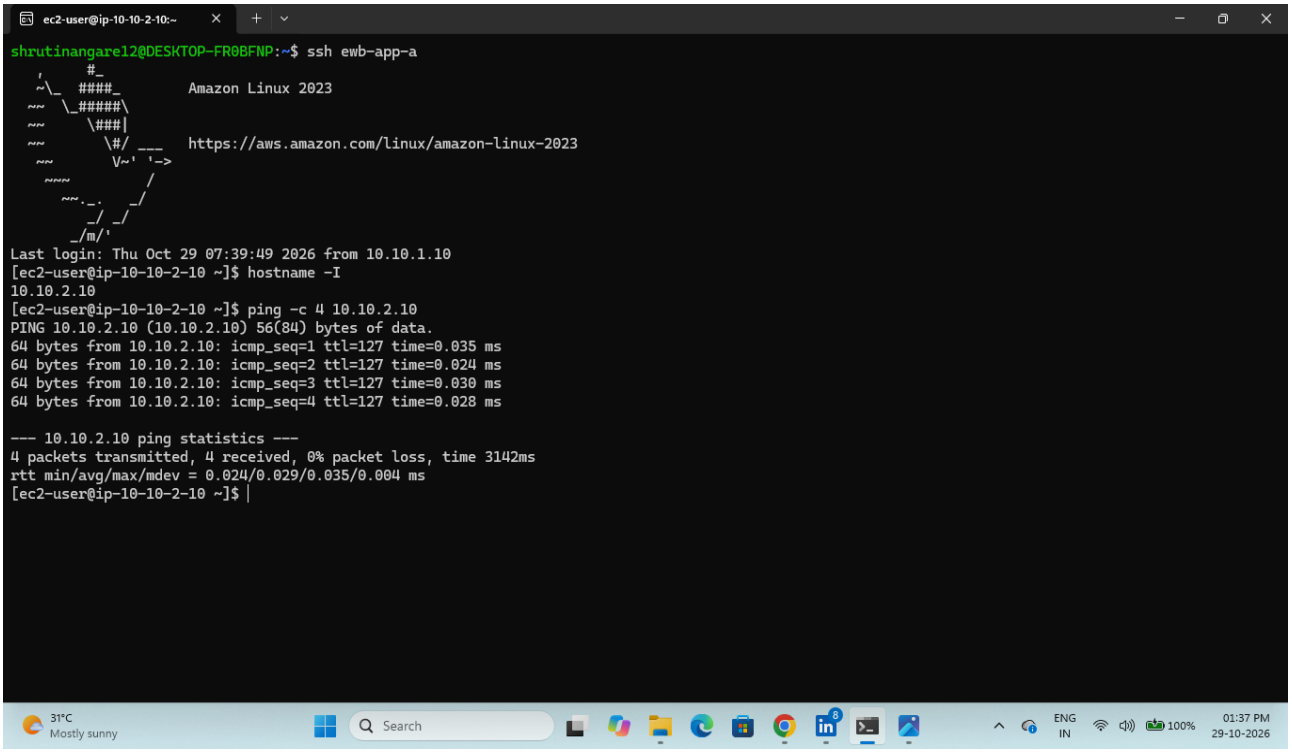
- Baseline (Before peering)



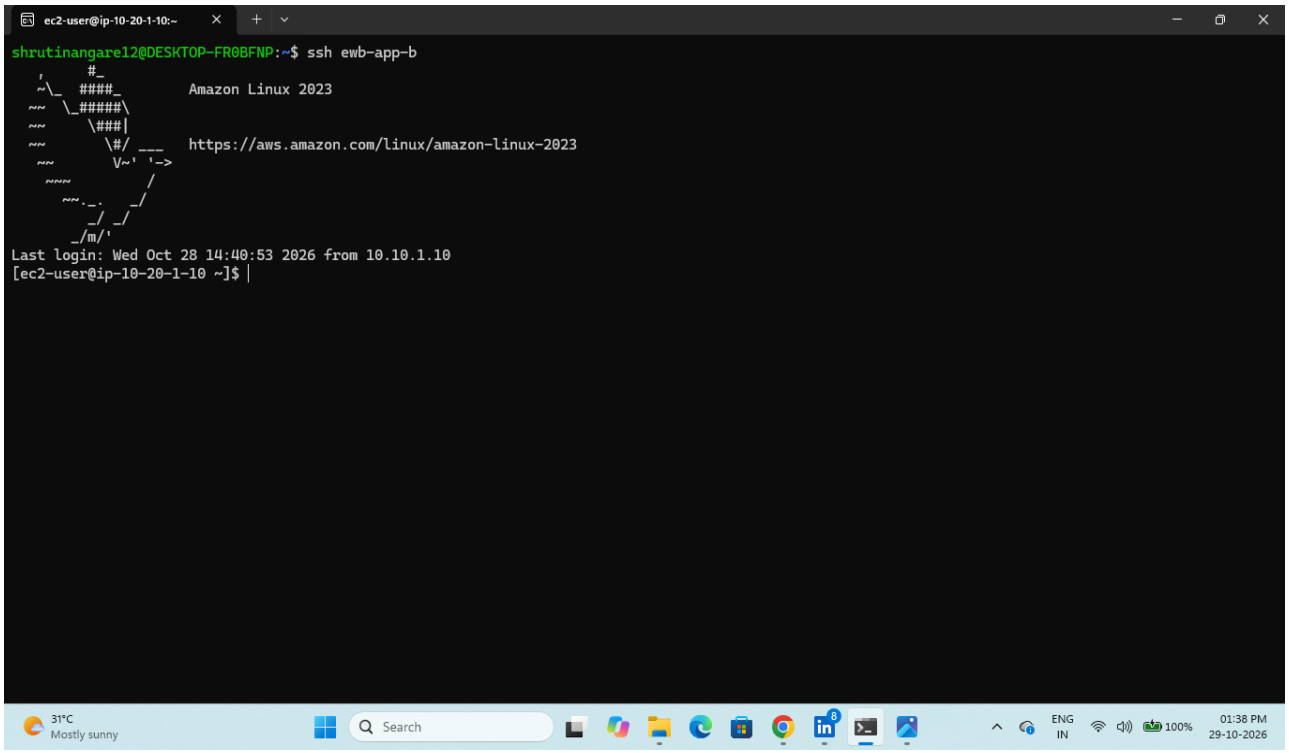
- Forward Path



❖ Reverse Path

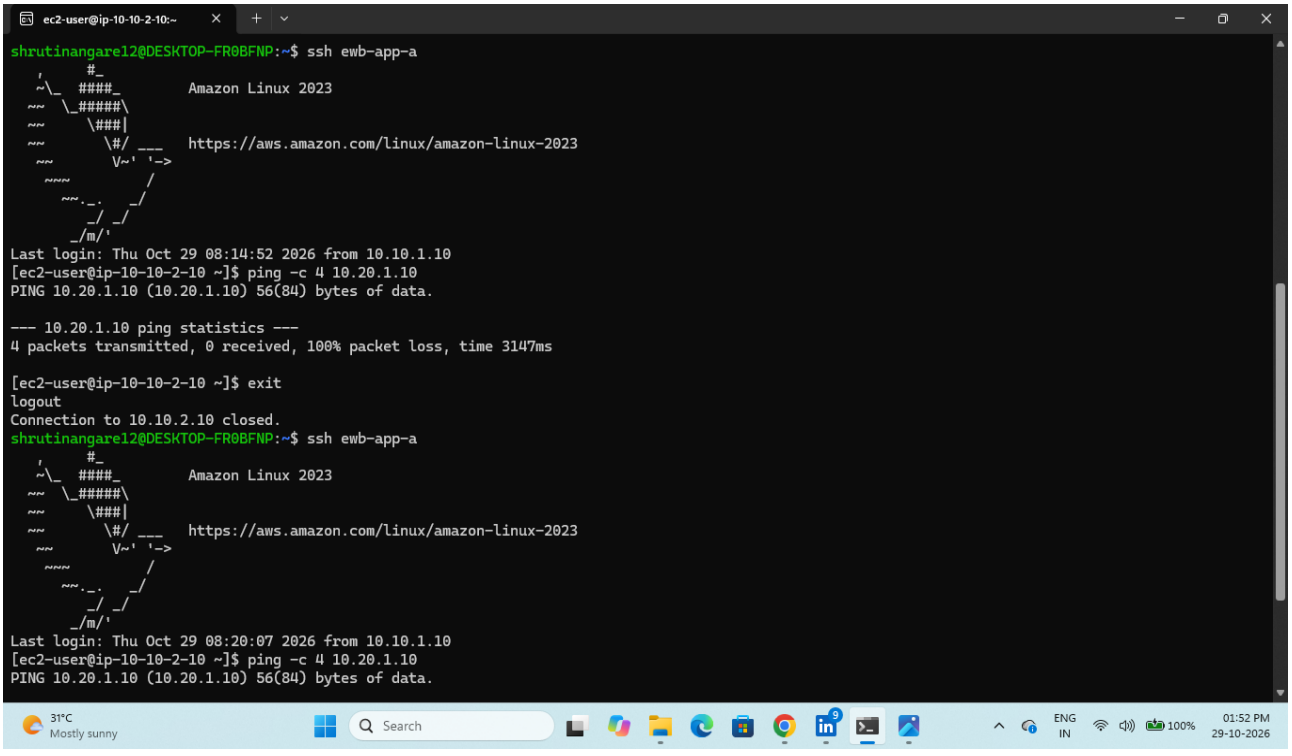


❖ Private SSH



❖ CONTROLLED ICMP FAILURE AND RECOVERY

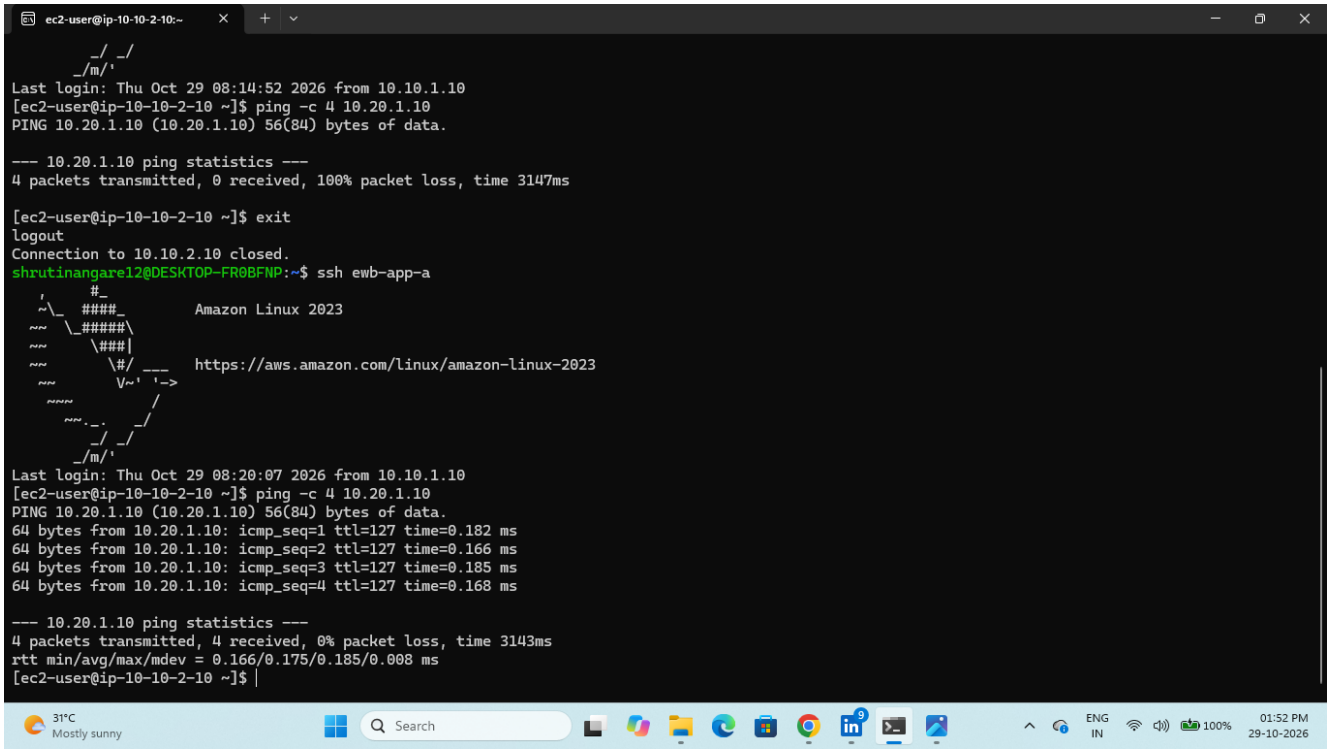
- When ICMP rule removed from App-B



Controlled ICMP Failure: The ICMP Echo Request rule allowing App-A (10.10.2.10) to reach App-B (10.20.1.10) was temporarily removed from App-B's security group. As a result,

ping requests were blocked and no ICMP replies were received. The VPC peering and routes remained unchanged.

- When ICMP rule assign to App-B



ICMP Recovery: The exact ICMP Echo Request rule was restored in App-B's security group. Ping from App-A to App-B successfully resumed, confirming that the connectivity was restored.

❖ In this project I have learned,

- How to Create and configure VPCs, subnets, route tables, and Internet Gateway.
- Connect two VPCs using VPC Peering.
- Configure routes and security groups for private communication.
- Use a bastion host to securely access private EC2 instances.
- Test connectivity using SSH and ping.
- Troubleshoot network connectivity and perform ICMP failure and recovery testing.
- Understand the importance of forward and return routes in network communication.